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Started	Wednesday, 21 October 2026, 8:37 PM
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Duration	3 mins 22 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. When the item is not the array at all

Question **2**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The lower middle element is chosen as the middle element ✗

b. The higher middle element is chosen as the middle element

c. The search stops

d. The middle element is chosen randomly

Question **3**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist

b. The search continues in the right sublist

c. The search ends successfully ✓

d. The list is sorted

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(1)
c. O🥎
d. O(log n) ✓

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It requires the list to be sorted ✓
c. It is slower than linear search for small lists
d. It does not work with negative numbers

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 90 and 100
b. 89 and 94
c. 94 and 99
d. 90 and 99 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is sorted ✓
b. When the list is very small
c. When the list contains duplicate elements
d. When the list is unsorted

Question **8**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Selection

b. Bubble ✓

c. Insertion

d. Complexity

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list

b. At the beginning of the list

c. In the left sublist ✓

d. In the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. log n

b. n ✓

c. n/2

d. 1

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Binary search algorithm is not efficient when the data elements more than 1500 ✓

b. Must use a sorted array

c. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

d. There must be a mechanism to access middle element directly

Question **12**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Linear search ✓

b. Hash search

c. Both (1) & (3)

d. Binary search

Question **13**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By sorting the list first

b. By comparing each element in the list sequentially ✓

c. By dividing the list into halves

d. By using a hash function

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓

b. Linear search is more efficient than binary search on large lists.

c. Linear search requires the list to be sorted.

d. Linear search divides the list into halves.

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n-1
d. n ✓